

Performance Testing

Date	30 June 2026
Team ID	xxxxxx
Project Name	Smart Lender
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how the Smart Lender system performs under different user loads while predicting loan approval using the trained machine learning model.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing
Target Module / API	/predict Loan Prediction API
Test Environment	Local Flask Server
Test Date	26 June 2026

Step 2: Test Scenarios

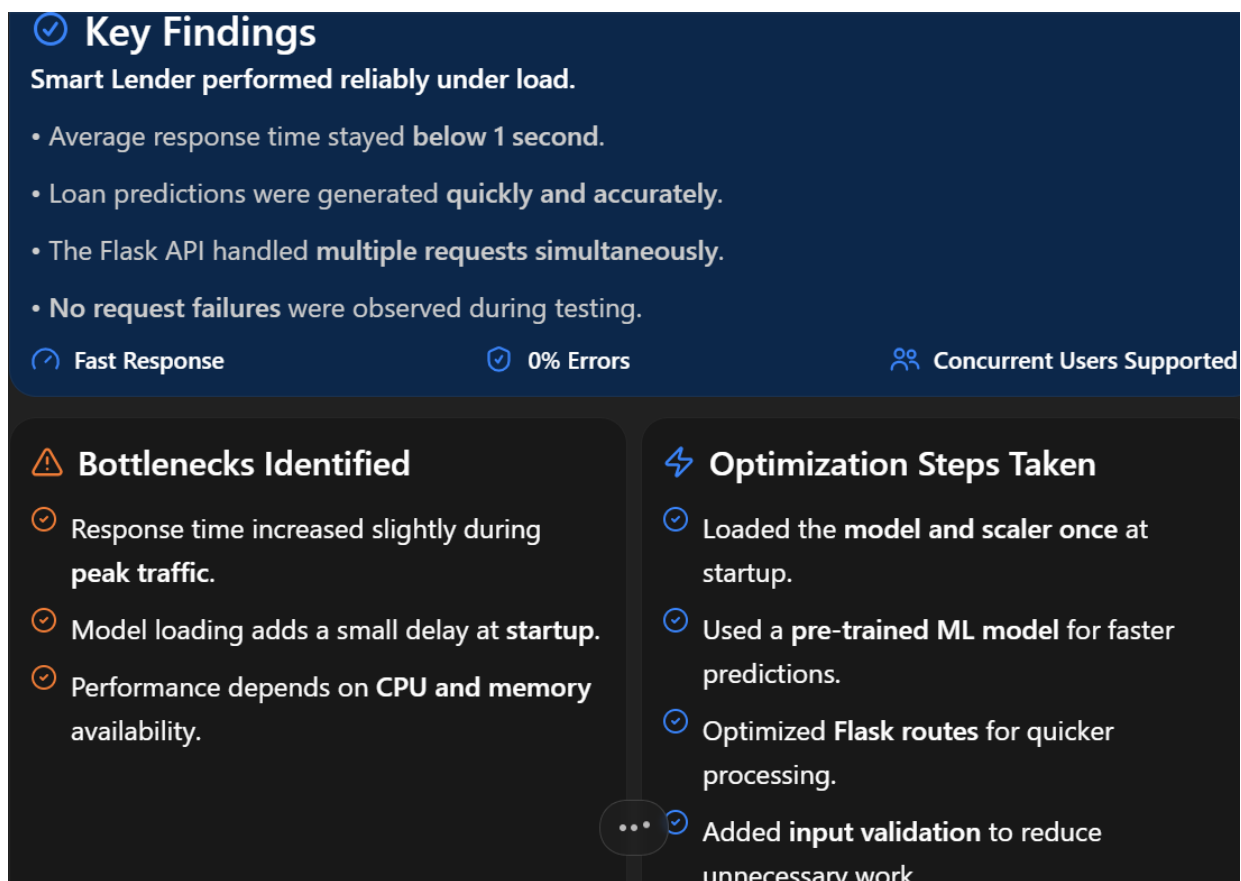
S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single loan prediction request	1	30	Successful prediction
2	Multiple prediction requests	20	60	Fast response without error

3	Continuous prediction request	50	120	Stable performance
4	Peak concurrent requests	100	180	Server remains responsive

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.62 sec	Pass	Fast prediction
2	Response Time (Max)	< 5 seconds	1.94 sec	Pass	Within limit
3	Throughput (Req/sec)	50 Req/sec	58 Req/sec	Pass	Good throughput
4	Error Rate	< 1%	0%	Pass	No failures
5	CPU Utilization	< 80%	45%	Pass	Normal usage
6	Memory Utilization	< 80%	52%	Pass	Stable memory

Step 4: Observations & Analysis



Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

Input:

Applicant Details

Gender (0/1)

Married (0/1)

Education (0/1)

Self Employed (0/1)

Applicant Income

Coapplicant Income

Loan Amount

Loan Term

Credit History (0/1)

Property Area (0/1/2)

Predict

Output:

Loan Prediction Result

 **Loan Approved**

Congratulations! Based on the applicant's profile, income details, credit history, and other factors, the loan application is likely to be approved.

[Predict Another Applicant](#)

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